

1. Administrative & Study Identification

Official Title: A Phase 3, Two-stage, Randomized, Multi-center, Controlled, Open-label Study Comparing Iberdomide Maintenance to Lenalidomide Maintenance Therapy After Autologous Stem Cell Transplantation (ASCT) in Participants With Newly Diagnosed Multiple Myeloma (NDMM) **Brief Study Title:** A Study to Compare Iberdomide Maintenance Versus Lenalidomide Maintenance Therapy Following Autologous Stem Cell Transplant in Participants With Newly Diagnosed Multiple Myeloma **Short Title/Acronym:** EXCALIBER-Maintenance **Protocol Number:** IM048-022 **NCT Number:** NCT05827016 **Posting ID:** 122086711069803034 **Sponsor/Company Name:** Bristol Myers Squibb **Investigational Product:** Iberdomide (CC-220) vs. Lenalidomide **Phase of Development:** PHASE3 **Trial Status:** RECRUITING (Currently recruiting participants) **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To compare the effectiveness, specifically Progression-Free Survival (PFS), of iberdomide maintenance to lenalidomide maintenance therapy after autologous stem cell transplantation (ASCT) in participants with newly diagnosed multiple myeloma (NDMM). **Secondary Objectives:** To evaluate safety, Minimal Residual Disease (MRD) negativity in patients with a Complete Response (CR) at 12 months, Overall Survival (OS), time to progression, Health-Related Quality of Life (HRQoL), and to determine the recommended iberdomide dose for Stage 2.

2.2 Background & Rationale While lenalidomide has established itself as the standard of care for maintenance therapy following ASCT in newly diagnosed multiple myeloma, disease progression remains a challenge. Iberdomide is a novel, potent oral cereblon E3 ligase modulator (CELMoD) with enhanced direct antitumor and immune stimulatory effects compared to lenalidomide. This study is being conducted to determine if the optimized specificity and potency of iberdomide can provide superior therapeutic benefit, deepen clinical responses, and extend progression-free survival in the post-ASCT maintenance setting.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Active Comparator (Lenalidomide) **Blinding:** Open-label (None) **Randomization:** Randomized (Parallel Assignment, Two-stage) **Phase**

Requirement: Trial must be in PHASE3 (Criteria Type: PHASE_REQUIREMENT)

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 1216 participants **Number of Sites:** Multi-center (Global) **Estimated Study Start:** 22-Jun-2023 **Estimated Primary Completion:** 15-Mar-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Confirmed diagnosis of symptomatic multiple myeloma (MM) with an Eastern Cooperative Oncology Group (ECOG) performance status score of 0, 1, or 2. 2. Received 3 to 6 cycles of induction therapy including a proteasome inhibitor (PI) and immunomodulatory drug (IMiD) (e.g., VTd, RVd, VCd) with or without a CD38 monoclonal antibody, followed by a single or tandem autologous stem cell transplantation (ASCT). 3. Achieved at least a partial response (PR) after ASCT within 12 months (single transplant) or 15 months (tandem transplant) from the initiation of induction therapy.

4.2 Exclusion Criteria 1. Progressive disease or clinical relapse (as defined by IMWG response criteria) following ASCT with or without consolidation, or non-responsiveness to primary therapy. 2. Smoldering myeloma, solitary plasmacytoma, or nonsecretory myeloma. 3. Known central nervous system/meningeal involvement of MM, or a prior history of malignancies (other than MM) unless free of the disease for $\geq$ 5 years.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Experimental Arms (A1, A2, A3):** Ixerdomide (at Dose 1, Dose 2, or Dose 3) taken daily for the first 21 days of each 28-day cycle. * **Active Comparator Arm (B):** Lenalidomide (Trade Name: Revlimid; ATC Code: L04AX04; EMA Info: [Product Information](#)) taken daily (starting at 10 mg, with a potential increase to 15 mg after 3 cycles depending on tolerability and response).

Route of Administration: Oral **Duration of Treatment:** Up to approximately 4 years, or until disease progression/unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Post-stem cell transplant consolidation therapy is permitted prior to maintenance. Standard supportive care according to institutional guidelines. **Prohibited:** Concurrent investigational agents or other systemic anti-myeloma therapies not specified in the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Physical Exam, ECG, Blood/Urine Labs, Disease Status Evaluation (IMWG criteria)
Baseline	Day 0	Randomization, First Dose of Maintenance Therapy
Treatment Cycles	Day 1 of 28-Day Cycle	Safety Labs, Efficacy Assessment (Blood/Urine for M-protein), IP Accountability, AE Monitoring
Interim (Month 12)	Month 12 [Window]	Bone Marrow Aspirate for MRD Assessment (for patients in $\geq$ CR)
End of Treatment	Post-Progression	Final Vital Signs, Exit Interview, Safety Follow-up, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS) – the time from randomization to the first documentation of objective disease progression or death due to any cause. Evaluated over a timeframe of up to 6 years. **Measurement Method:** Clinical and laboratory evaluation per International Myeloma Working Group (IMWG) criteria.

7.2 Secondary & Exploratory Endpoints * Minimal Residual Disease (MRD) negativity status in participants who achieve CR or better at 12 months ($\pm$ 3 months) post-maintenance initiation. * Overall Survival (OS). * Recommended ibendomide dose for Stage 2. * Time to progression (TTP) and Duration of Response (DoR). * Health-Related Quality of Life (HRQoL) and overall Safety Profile.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring at each cycle visit, with specific attention to hematologic toxicities (e.g., neutropenia, thrombocytopenia) and infections, typical of CELMoDs and IMiDs. **Serious Adverse Events (SAEs):** Must be reported to the sponsor within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee will regularly review unblinded safety and efficacy data to ensure patient safety and study integrity.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy analysis (PFS). Safety Set for all participants who receive at least one dose of the study drug for AE/SAE evaluations. **Sample Size Justification:** $N=1216$ appropriately powered to detect a statistically significant improvement in PFS

for iberdomide compared to lenalidomide maintenance. **Handling of Missing Data:** Standard methodologies per oncology trial guidelines; censoring for PFS will occur at the last adequate disease assessment for patients without documented progression.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring visits to ensure protocol adherence, accurate data transcription, and safety reporting. **Audit Trail:** eCRF locking and data

clarification forms via standard quality audit mechanisms. **Source Level:** 5

11. Study Identifiers & Governance

Primary ID: IM048-022 **Registry Number:** NCT05827016 **Posting ID:** 122086711069803034 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** EXCALIBER-Maintenance

12. Clinical Framework (Multiple Myeloma Specific)

Disease Area / Primary Condition: Multiple Myeloma **Preferred UMLS Name:** Multiple Myeloma Treatment **Semantic Type:** Neoplastic Process **Disease Definition:** A bone marrow-based plasma cell neoplasm characterized by a serum monoclonal protein and skeletal destruction with osteolytic lesions, pathological fractures, bone pain, hypercalcemia, and anemia. Clinical variants include non-secretory myeloma, smoldering myeloma, indolent myeloma, and plasma cell leukemia. (WHO, 2001) **Diagnosis Threshold:** Symptomatic Multiple Myeloma achieving at least PR after induction and ASCT **Medical Methodology:** Post-ASCT Maintenance Therapy **Optimization Target:** Progression-Free Survival (PFS) and MRD Negativity **Clinical Coding Mapping:** * **MeSH Code:** D009101 * **HPO Code:** HP:0006775 * **SNOMED ID:** 109989006 * **SNOMED Hierarchy:** * 269475001 | Malignant neoplasm of lymphoid, hematopoietic and/or related tissue (disorder) * 415111003 | Plasma cell neoplasm (disorder) * 1162576007 | Plasma cell myeloma (morphologic abnormality) * 64572001 | Disease (disorder)

13. Study Procedures & Methodology

Management Pathway: Standard maintenance clinical pathway post-ASCT for NDMM **Education Protocol (HCP):** AE profile training for CELMoDs vs IMiDs (e.g., hematologic toxicity management)

Education Protocol (Patient): Oral medication adherence training (21-day on/7-day off schedule for iberdomide vs. continuous for

lenalidomide) **Quality Audit Mechanism:** Centralized laboratory assessments for MRD and IMWG response tracking

14. Observation & Follow-Up Schedule

Total Duration: Up to 6 years (Estimated Completion: 2036) **Onsite Visit Cadence:** Every 28 days (Day 1 of each cycle) **Remote Monitoring Frequency:** Standard protocol intervals **Checklist Review Interval:** Cycle-by-cycle disease evaluation (IMWG criteria)

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				